

[FACE RECOGNITION ON SKEWED UAV IMAGES]

SETUP GUIDE

Clone, Configure and Run on Any Windows Laptop

Step-by-step instructions written so that even someone who has never seen this project before can get it running successfully.

Follow every step in order. Do not skip any step.

PLATFORM

Windows 10 / 11

PYTHON

3.10 or 3.11

FRAMEWORK

FastAPI + InsightFace

github.com/TENSEI3011/face-recognition-skewed-images

Do NOT skip any step. Follow from top to bottom.

What This Guide Covers

This guide walks you through setting up and running the Face Recognition UAV project on a brand-new Windows laptop. It is written for someone who has never used this project before. Follow every step in order and do not skip anything.

What the Laptop Needs BEFORE You Start

Requirement	Version / Details	How to Check
Python	3.10 or 3.11 (NOT 3.12 or 3.13)	Open CMD: python --version
Git	Any version	Open CMD: git --version
Internet	Required for first setup	Or transfer files via USB
RAM	Minimum 8 GB	Task Manager > Performance
Disk Space	Minimum 5 GB free	Windows Explorer
Browser	Chrome or Edge	Already installed on Windows

[WARNING] Python Version is Critical

Python 3.13 causes dlib-bin to FAIL during installation.

Only Python 3.10 or 3.11 are guaranteed to work with all dependencies.

If the laptop already has Python 3.13, install Python 3.11 from python.org alongside it.

[NOTE] Fully Offline After First Setup

Run `python offline_setup.py` once on any machine with internet.

It downloads all models and fonts. After that, the system runs with ZERO internet.

For air-gapped / defence deployment: copy the full project folder + `~/insightface/models/buffalo_l/` via USB.

STEP 1 - INSTALL PYTHON 3.11

1. Open Chrome or Edge on the scientist laptop.
2. Go to: <https://www.python.org/downloads/release/python-3119/>
3. Scroll down to the Files section. Click: Windows installer (64-bit)
4. Run the downloaded .exe file. On the FIRST installer screen:

--> TICK THE CHECKBOX: 'Add Python.exe to PATH' <-- Do NOT miss this!

5. Click 'Install Now'. Wait for it to complete. Click Close.

Verify Python installed correctly:

Press Windows key, type cmd, press Enter to open Command Prompt. Type:

Command Prompt

```
python --version
```

You should see: Python 3.11.9 (or any 3.11.x number)

[COMMON ERROR] python is not recognized as an internal or external command

Python was not added to PATH during installation.

Uninstall Python (Windows Settings > Apps), re-run the installer, and make sure you tick 'Add Python.exe to PATH' on the first screen.

STEP 2 - INSTALL GIT

1. Open browser and go to: <https://git-scm.com/download/win>
2. Click '64-bit Git for Windows Setup' to download.
3. Run the installer. Click Next on every screen (all defaults are fine).
4. Click Install, then Finish.

Verify Git is installed:

Command Prompt

```
git --version
```

You should see: git version 2.x.x windows x

STEP 3 - CLONE THE GITHUB REPOSITORY

Open Command Prompt (Windows key > type cmd > Enter). Type these commands:

Command Prompt - type each line and press Enter

```
cd Desktop
git clone https://github.com/TENSEI3011/face-recognition-skewed-images.git
cd face-recognition-skewed-images
```

A folder called 'face-recognition-skewed-images' will appear on the Desktop.

[TIP] Slow Internet or No Internet?

Copy the entire 'face-recognition-skewed-images' folder from Lucky's laptop to the USB drive and paste it on the scientist's Desktop.

Then open CMD and type: `cd Desktop\face-recognition-skewed-images`

STEP 4 - CREATE A VIRTUAL ENVIRONMENT

A virtual environment is an isolated Python workspace. Make sure you are inside the project folder (from Step 3).
Run:

Command Prompt - run inside the project folder

```
python -m venv venv

venv\Scripts\activate
```

After the second command, you will see (venv) at the start of the prompt:

What the prompt looks like when venv is active

```
(venv) C:\Users\...\face-recognition-skewed-images>
```

(venv) MUST be visible before you continue to the next step

[WARNING] Always activate (venv) every time you open a new CMD window

Every time you close and reopen Command Prompt, you must run
venv\Scripts\activate again before running any python commands.
If you forget, you will see ModuleNotFoundError errors.

[COMMON ERROR] venv\Scripts\activate gives an execution policy error

This can happen in PowerShell. Run this command once:
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
Press Y then Enter. Then try activate again.
Or use Command Prompt (cmd.exe) instead of PowerShell - it has no such restriction.

STEP 5 - INSTALL ALL PYTHON PACKAGES

Make sure (venv) is showing in your prompt. Then run these two commands:

Command Prompt (venv must be active)

```
pip install --upgrade pip
pip install -r requirements.txt
```

This downloads and installs everything: ArcFace, FAISS, OpenCV, scikit-learn,

FastAPI, dlib, and many more. It takes 10 to 20 minutes on first run.

Hundreds of lines of output are completely normal. Wait for 'Successfully installed'.

[COMMON ERROR] dlib-bin install failed - Microsoft Visual C++ 14.0 is required

Run these commands in order:

```
pip install cmake
pip install dlib
```

Then try again: pip install -r requirements.txt

[COMMON ERROR] ERROR: No matching distribution found (Python 3.12 or 3.13)

Your Python version is incompatible. Check: `python --version`

If it shows 3.12 or 3.13, install Python 3.11 from python.org (Step 1).

Then delete the venv folder and run:

```
py -3.11 -m venv venv
```

```
venv\Scripts\activate
```

```
pip install -r requirements.txt
```

STEP 6 - DOWNLOAD MODELS AND FONTS (ONE-TIME INTERNET SETUP)

Run this single command to download EVERYTHING needed for offline operation: dlib landmark model, InsightFace buffalo_l (ArcFace + SCRFD), and all 9 UI fonts.

Command Prompt (venv active, internet required)

```
python offline_setup.py
```

This downloads and caches:

- dlib 68-point landmark model (~100 MB) -> models/ in project folder
- InsightFace buffalo_l model (~500 MB) -> C:\Users\<name>\.insightface\models\buffalo_l\
- All 9 UI fonts (.ttf) -> web/frontend/fonts/ (no more internet needed for UI)

After this completes, the system runs FULLY OFFLINE.

[TIP] Slow Internet or No Internet? Copy via USB Instead

Prepare these items on Lucky's laptop and copy to USB:

1. models\shape_predictor_68_face_landmarks.dat (~100 MB)
2. C:\Users\lucky\.insightface\models\buffalo_l\ (~500 MB)
3. web\frontend\fonts\ (all .ttf files, ~1.5 MB)
4. data\gallery\ (enrolled identity photos)
5. results\baseline\models\ (trained SVM + FAISS)

Paste each to the same relative location on the scientist laptop.

How to Copy the buffalo_l InsightFace Folder (Most Important):

- On Lucky's laptop, open File Explorer
- In the top address bar, type this path and press Enter:

```
C:\Users\lucky\.insightface\models\
```

Type this in

If you cannot see the .insightface folder, click View tab > tick 'Hidden items'

- Copy the entire 'buffalo_l' folder to your USB drive
- On the scientist's laptop, open File Explorer and navigate to C:\Users\<name>\
- Create a new folder '.insightface', inside it create a folder 'models'
- Paste the 'buffalo_l' folder into: C:\Users\<name>\.insightface\models\

STEP 7 - CREATE THE .ENV CONFIGURATION FILE

The .env file stores the secret key and database connection. Run:

Command Prompt (inside the project folder)

```
copy .env.example .env
```

Then open the file in Notepad to edit it:

Command Prompt

```
notepad .env
```

The file will open in Notepad. Edit it to look like this (choose ONE MongoDB option):

.env file - save with Ctrl+S after editing

```
JWT_SECRET_KEY=face-recognition-uav-secret-key-2025

# OPTION A: Local MongoDB (recommended for offline / defence use)
MONGO_URI=mongodb://localhost:27017

# OPTION B: MongoDB Atlas (cloud - remove # to use)
#MONGO_URI=mongodb+srv://user:password@cluster.mongodb.net/

# OPTION C: No MongoDB (local disk only - comment out MONGO_URI)
#MONGO_URI=

MONGO_DB_NAME=facerecog_db
ENV=development

# --- Passive Anti-Spoofing (6-signal fusion) ---
LIVENESS_ENABLED=True          # Set False to disable the check
LIVENESS_THRESHOLD=0.45        # 0.0-1.0: lower = more permissive

# --- Active Blink Challenge (webcam / live mode only) ---
BLINK_ENABLED=True             # Set False to skip blink challenge
BLINK_EAR_THRESHOLD=0.25       # Eye Aspect Ratio below this = blink
BLINK_TIMEOUT_SEC=7.0          # Seconds user has to blink
BLINK_REQUIRED_COUNT=1         # Number of blinks required
BLINK_CONSEC_FRAMES=2          # Consecutive frames below EAR = 1 blink

# --- Recognition Threshold ---
FAISS_THRESHOLD=0.35           # Cosine similarity threshold (0.35 tuned for UAV)
```

Save the file (Ctrl+S) and close Notepad.

[TIP] Recommended for Defence / Offline Use: Local MongoDB

Install MongoDB Community Server (free) from mongodb.com/try/download/community

After installing, start it with: `mongod --dbpath C:\data\db`

Then set: `MONGO_URI=mongodb://localhost:27017` in your `.env` file.

This gives you full audit log, watchlist, and gallery features with NO internet.

[TIP] No MongoDB at All? Use Disk-Only Mode

Remove the `MONGO_URI` line from `.env` entirely.

Gallery images are stored as files in `data/gallery/`.

Audit log and watchlist are unavailable, but identification still works.

STEP 8 - START THE WEB SERVER

Make sure (venv) is active and you are in the project folder. Run this command:

Command Prompt - keep this window open while using the system

```
python -m uvicorn web.backend.main:app --host 0.0.0.0 --port 8000 --reload
```

Wait 20 to 30 seconds. If everything is set up correctly, you will see:

Expected output when everything is working correctly

```
INFO:      Uvicorn running on http://0.0.0.0:8000
[FaceDetector] SCRFD (buffalo_l) loaded successfully.
[ArcFaceExtractor] Loaded model: buffalo_l
[SVMClassifier] Loaded from results/baseline/models/svm_classifier.pkl
[FAISSMatcher] Loaded: 75 vectors, 3 identities, threshold=0.35
[LivenessService] Passive liveness detector initialised (threshold=0.45, 6-signal fusion)
[LivenessService] BlinkDetector initialised -- status: OK
INFO:      Application startup complete.
```

Now open Chrome or Edge and go to this address:

Browser address bar

```
http://localhost:8000
```

The Face Recognition dashboard will appear in the browser. The system is ready.

[COMMON ERROR] Website shows 'Pipeline not loaded'

The trained model files are missing.

Solution A: Copy `results\baseline\models\` from Lucky's laptop via USB (Step 6).

Solution B: Run this command to retrain from scratch (takes 20-30 mins):

```
python experiments\run_baseline.py
```

(Make sure gallery photos are in `data\gallery\` before running this.)

STEP 9 - ENROLL IDENTITIES AND RETRAIN

SKIP THIS STEP if you already copied results\baseline\models\ via USB in Step 6.

If models are missing OR you want to add new people, use the Gallery page:

Option A: Enroll via Web UI (recommended):

- Open <http://localhost:8000/gallery> in your browser
- Enter the person's name and click 'Upload Images' or drag a video file
- Supported formats: JPG, PNG, BMP (images) and MP4, MOV, AVI (video)
- For video: frames are sampled every 15 frames; blurry frames are skipped automatically
- Model retraining starts AUTOMATICALLY after each upload (no manual click needed)
- First retrain takes ~15 seconds. After that: ~2 seconds thanks to embedding cache

Option B: Retrain with GridSearchCV (best accuracy, takes ~5 minutes extra):

After your gallery is complete, run a GridSearch retrain to find the best SVM settings:

GridSearchCV retrain - improves Rank-1 accuracy by 5-15%

```
# Via browser at http://localhost:8000/docs -> POST /api/pipeline/retrain
# Set use_grid_search=true in the request body

# OR via command line:
curl -X POST "http://localhost:8000/api/pipeline/retrain?use_grid_search=true"
```

Option C: Full Baseline Experiment (most thorough, 20-30 minutes):

Command Prompt (venv active) - trains SVM, FAISS, and generates accuracy reports

```
python experiments\run_baseline.py
```

[WARNING] Gallery Photos Must Be Present for Retraining

data\gallery\ must contain identity subfolders with photos.

Copy from Lucky's laptop if empty. Use at least 10-30 photos per person for best accuracy.

Blurry / dark images are automatically rejected during enrollment.

STEP 10 - USING THE SYSTEM

Page	URL	What It Does
Dashboard	http://localhost:8000/	System status and metrics overview
Identify	http://localhost:8000/identify	Upload photo - passive liveness (6-signal) checked automatically
Gallery	http://localhost:8000/gallery	Enroll via images OR video files (MP4/MOV/AVI) - auto-retrains
Demo	http://localhost:8000/demo	Upload video or webcam - blink challenge active in webcam mode
Results	http://localhost:8000/results	Experiment accuracy graphs (CMC, ROC)
Analytics	http://localhost:8000/analytics	System usage stats
Audit Log	http://localhost:8000/audit	All identification events (incl. spoof / blink-fail detections)
Watchlist	http://localhost:8000/watchlist	Alert watchlist management
Config	http://localhost:8000/config	Threshold, pipeline settings, anti-spoofing + blink toggle
API Docs	http://localhost:8000/docs	Full REST API reference

Identify a Face in a Photo:

- Go to <http://localhost:8000/identify> in your browser
- Click 'Choose File' and select any .jpg or .png photo containing a face
- Click 'Identify'
- The system shows the person's name, confidence score, and a box around the face

Enroll a New Person from Video:

- Go to <http://localhost:8000/gallery>
- Enter the person's name and upload a 10-30 second MP4 or MOV video clip
- Set frame interval to 15 (default) ? captures ~2 faces per second
- Click Upload. Only sharp, clear frames are stored (blurry ones are rejected)
- Click 'Retrain from Gallery' to rebuild the model with the new identity

Process a Video File:

- Go to <http://localhost:8000/demo>
- Click the 'Upload Video' tab
- Click to upload and select a .mp4 video
- Adjust 'Process every N frames' slider (default 6 = fast; lower = more accurate but slower)
- Click 'Process Video' and wait for processing
- Download the output video with face names labelled on each detected person

To Stop the Server:

Go to the Command Prompt window where uvicorn is running and press Ctrl+C.

TROUBLESHOOTING - Common Errors and Their Fixes

[COMMON ERROR] python is not recognized as an internal or external command

Python is not installed or not in PATH.

Re-install Python 3.11 from python.org.

On the first installer screen, tick 'Add Python.exe to PATH' before clicking Install.

[COMMON ERROR] No module named 'web'

You are running the uvicorn command from the wrong folder.

Make sure you first cd into the project folder:

```
cd Desktop\face-recognition-skewed-images
```

Then run the uvicorn command again.

[COMMON ERROR] Website shows 'Pipeline not loaded'

The trained SVM and FAISS model files are missing.

Copy results\baseline\models\ from Lucky's laptop via USB (Step 6).

OR run: `python experiments\run_baseline.py` to retrain (needs gallery photos).

[COMMON ERROR] shape_predictor_68_face_landmarks.dat not found

The dlib facial landmark model file is missing.

Copy models\shape_predictor_68_face_landmarks.dat from Lucky's laptop via USB.

[COMMON ERROR] InsightFace is downloading models very slowly at startup

The buffalo_I model folder is not in the right location.

Copy the buffalo_I folder from Lucky's laptop (Step 6, item 2).

Place it at: `C:\Users\<scientist-name>\.insightface\models\buffalo_I\`

[COMMON ERROR] Port 8000 is already in use

Another program is using port 8000. Use port 8001 instead:

```
python -m uvicorn web.backend.main:app --host 0.0.0.0 --port 8001 --reload
```

Then open `http://localhost:8001` in the browser.

[COMMON ERROR] dlib install failed: Microsoft Visual C++ 14.0 is required

Install Microsoft C++ Build Tools:

1. Search 'Visual Studio Build Tools' download in browser
2. Download and run vs_BuildTools.exe
3. Select 'Desktop development with C++' and click Install
4. After it finishes, run: `pip install cmake && pip install dlib`
5. Then: `pip install -r requirements.txt`

[COMMON ERROR] PowerShell: venv\Scripts\activate gives execution policy error

Run this in PowerShell first:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

Press Y then Enter. Then run `venv\Scripts\activate` again.

Alternatively, use Command Prompt (`cmd.exe`) instead of PowerShell.

QUICK START CHEAT SHEET - Print This Page and Keep It Handy

Every Time You Open a New Terminal Window, Run These 3 Commands:

3 commands to start the server

```
cd Desktop\face-recognition-skewed-images
venv\Scripts\activate
python -m uvicorn web.backend.main:app --host 0.0.0.0 --port 8000 --reload
```

Then open Chrome and go to: <http://localhost:8000>

All One-Time First-Day Setup Commands (Run Once Only):

Complete first-time setup from scratch

```
# 1. Clone the repository
cd Desktop
git clone https://github.com/TENSEI3011/face-recognition-skewed-images.git
cd face-recognition-skewed-images

# 2. Create virtual environment
python -m venv venv
venv\Scripts\activate

# 3. Install all packages (takes 10-20 minutes)
pip install --upgrade pip
pip install -r requirements.txt

# 4. Download ALL models + fonts (one-time internet step)
python offline_setup.py

# 5. Create configuration file
copy .env.example .env
notepad .env <- set MONGO_URI=mongodb://localhost:27017

# 6. Start the server
python -m uvicorn web.backend.main:app --host 0.0.0.0 --port 8000 --reload
```

USB Files Checklist - Prepare These Before Going (if no internet on target machine):

Item	File / Folder to Copy	Destination on New Laptop	Done?
1	models\shape_predictor_68_face_landmarks.dat	models\ in project folder	☐
2	C:\Users\lucky\.insightface\models\buffalo_1\	C:\Users\<name>\.insightface\models\buffalo_1\	☐
3	web\frontend\fonts\ (all .ttf files)	web\frontend\fonts\ in project folder	☐
4	data\gallery\ (all identity subfolders)	data\ in project folder	☐
5	results\baseline\models\ (SVM + FAISS)	results\baseline\ in project folder	☐